

Bilirubin, Direct

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate liver and biliary disease. Increased direct bilirubin occurs with biliary diseases, including both intrahepatic and extrahepatic lesions. Hepatocellular causes of elevation include hepatitis, cirrhosis, and advanced neoplastic states. Increased with cholestatic drug reactions, Dubin-Johnson syndrome, and Rotor syndrome. In the latter two syndromes, the level is usually <5 mg/dL.

Contraindications

Measurement of direct bilirubin is usually not necessary when the total bilirubin is <1.2 mg/dL.

Custom Additional Information

Theoretically, direct bilirubin should not be increased in hemolytic anemias, in which bilirubin increase should be in the indirect bilirubin fraction in the absence of complications. In practice, some increase in the direct fraction may be encountered in patients with hemolytic anemia in whom complications have not been proven. Some methods have shown the direct bilirubin to be spuriously high. This may be due to different concentrations of sodium nitrite, which may convert some of the unconjugated bilirubin to conjugated bilirubin.^{1,2} Direct bilirubin is the water soluble fraction. When increased in serum, bilirubin should become positive in the urine. Physiologic jaundice, occurring two to four days after birth, is due to lack of liver glucuronyl transferase.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	2 days
Refrigerated	3 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Range (mg/dL)
0 to 31 d	0.00–0.60
Children 1 month and older and adults	0.00–0.40

Storage Instructions

Refrigerate

Causes for Rejection

Gross hemolysis; improper labeling; gross lipemia

Footnotes

1. Chan KM, Scott MG, Wu TW, et al. Inaccurate values for direct bilirubin with some commonly used direct bilirubin procedures. *Clin Chem*. 1985 Sep; 31(9):1560-1563. PubMed 4028405

(<http://www.ncbi.nlm.nih.gov/pubmed/4028405>)

2. Mair B, Klempner LB. Abnormally high values for direct bilirubin in the serum of newborns as measured with the DuPont aca®. *Am J Clin Pathol*.

1987 May; 87(5):642-644. PubMed 3578139

(<http://www.ncbi.nlm.nih.gov/pubmed/3578139>)